

Module-3 Image Management & Registry Operations

Docker images

List and understand existing Docker images on your system.

→ docker images

```
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker/welcome-to-docker:latest	c4d56c24da4f	22.9MB	6.35MB	U
my-calc:latest	af45abf411da	91.8MB	25.8MB	U
param-demo1new:latest	532cf09e1d8f	202MB	43.5MB	U
redis:latest	7b6fb55d8b0a	225MB	55.4MB	

```
param@Params-MacBook-Air desktop %
```

Pull images from Docker Hub and inspect their details.

→ docker pull python

```
param@Params-MacBook-Air desktop % docker pull python
Using default tag: latest
latest: Pulling from library/python
4b75252fb560: Pull complete
1bd4defc8c5e: Pull complete
cace8fbd9245: Pull complete
bcc9a0bdacbc: Pull complete
642b703f20ff: Pull complete
bc8128ce97cc: Pull complete
7150cbe43a7c: Pull complete
184933270923: Download complete
4a00d37b54be: Download complete
Digest: sha256:151ab3571dad616bb031052e86411e2165295c7f67ef27206852203e854bcd12
Status: Downloaded newer image for python:latest
docker.io/library/python:latest
param@Params-MacBook-Air desktop %
```

→ docker inspect redis

```
param@Params-MacBook-Air desktop % docker inspect redis
[
  {
    "Id": "sha256:7b6fb55d8b0adcd77269dc52b3cffe5f59ca5d43dec3c90dbe18aacce7942e1",
    "RepoTags": [
      "redis:latest"
    ],
    "RepoDigests": [
      "redis@sha256:7b6fb55d8b0adcd77269dc52b3cffe5f59ca5d43dec3c90dbe18aacce7942e1"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-02-10T19:10:04.839782403Z",
    "Config": {
      "ExposedPorts": {
        "6379/tcp": {}
      },
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
      ],
      "Entrypoint": [
        "docker-entrypoint.sh"
      ],
      "Cmd": [
        "redis-server"
      ],
      "WorkingDir": "/data"
    },
    "Architecture": "arm64",
    "Variant": "v8",
    "Os": "linux",
    "Size": 53349888,
    "RootFS": {
      "Type": "layers",

```

Tag Docker images with custom names and versions.

→ `docker tag python python3`

```
param@Params-MacBook-Air desktop % docker tag python python3
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
<code>docker/welcome-to-docker:latest</code>	<code>c4d56c24da4f</code>	22.9MB	6.35MB	U
<code>my-calc:latest</code>	<code>af45abf411da</code>	91.8MB	25.8MB	U
<code>param-demo1new:latest</code>	<code>532cf09e1d8f</code>	202MB	43.5MB	U
<code>python3:latest</code>	<code>151ab3571dad</code>	1.63GB	421MB	
<code>python:latest</code>	<code>151ab3571dad</code>	1.63GB	421MB	
<code>redis:latest</code>	<code>7b6fb55d8b0a</code>	225MB	55.4MB	

```
param@Params-MacBook-Air desktop %
```

Remove unused or unwanted images to keep the system clean.

→ `docker rmi python3`

```
param@Params-MacBook-Air desktop % docker rmi python3
Untagged: python3:latest
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
<code>docker/welcome-to-docker:latest</code>	<code>c4d56c24da4f</code>	22.9MB	6.35MB	U
<code>my-calc:latest</code>	<code>af45abf411da</code>	91.8MB	25.8MB	U
<code>param-demo1new:latest</code>	<code>532cf09e1d8f</code>	202MB	43.5MB	U
<code>python:latest</code>	<code>151ab3571dad</code>	1.63GB	421MB	
<code>redis:latest</code>	<code>7b6fb55d8b0a</code>	225MB	55.4MB	

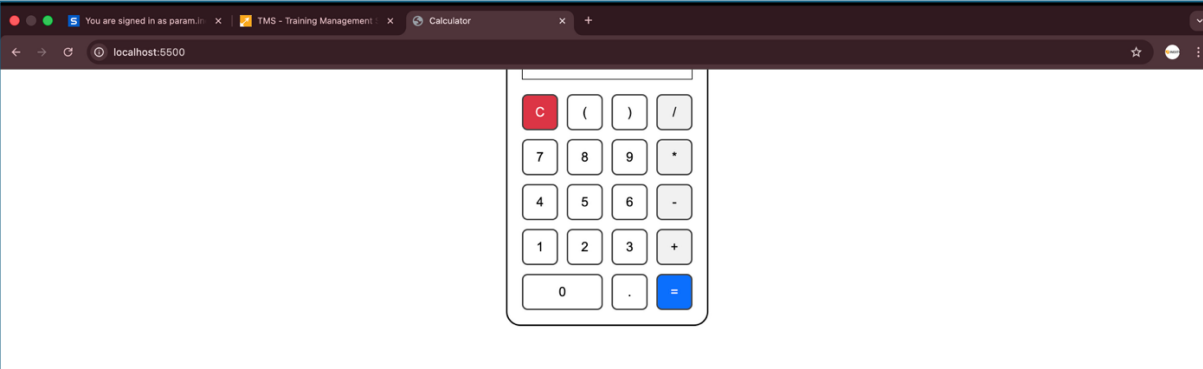
```
param@Params-MacBook-Air desktop %
```

Container Commit

Run a container from an existing image.

→ `docker run -d -p 5500:80 my-calc`

```
param@Params-MacBook-Air desktop % docker run -d -p 5500:80 my-calc
d889c9794bde01ff6528271ea823cbb04f2d83433deb8892cde6552c8a598dc7
param@Params-MacBook-Air desktop %
```



The screenshot shows a web browser window with the address bar set to `localhost:5500`. The page content is a simple calculator interface with a grid of buttons including digits 0-9, basic arithmetic operators (+, -, *, /), and a red 'C' button for clearing. Below the browser window, a terminal window displays the command `docker ps` and its output, showing the container `d889c9794bde` running the `my-calc` image with the command `"/docker-entrypoint..."`. The terminal also shows the container's status as 'Up About a minute' and its ports mapping.

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
d889c9794bde	my-calc	"/docker-entrypoint..."	About a minute ago	Up About a minute	0.0.0.0:5500->80/tcp, [::]:5500->80/tcp	heuri

```
param@Params-MacBook-Air desktop % docker ps
```

Make a small change inside the running container (for example, install a package or update a file).

→ `docker exec -it heuristic_davinci /bin/sh`

```
param@Params-MacBook-Air desktop % docker exec -it heuristic_davinci /bin/sh
/ # echo "Param was here" > test.txt
/ # ls
bin                etc                mnt                run                test.txt
dev                home              opt                sbin              tmp
docker-entrypoint.d lib               proc               srv               usr
docker-entrypoint.sh media             root               sys               var
/ # cat test.txt
Param was here
/ # exit
param@Params-MacBook-Air desktop %
```

Create a new Docker image from the modified container using the commit process.

→ `docker commit heuristic_davinci commit-container1`

```
param@Params-MacBook-Air desktop % docker commit heuristic_davinci commit-container1
sha256:cb245016a8d1a9f8298f9171771384df31e3751d9984500a34667917c784e306
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
commit-container1:latest	cb245016a8d1	91.9MB	25.8MB	
docker/welcome-to-docker:latest	c4d56c24da4f	22.9MB	6.35MB	U
my-calc:latest	af45abf411da	91.8MB	25.8MB	U
param-demo1new:latest	532cf09e1d8f	202MB	43.5MB	U
python:latest	151ab3571dad	1.63GB	421MB	U
redis:latest	7b6fb55d8b0a	225MB	55.4MB	

```
param@Params-MacBook-Air desktop % docker commit heuristic_davinci commit-container1
sha256:cb245016a8d1a9f8298f9171771384df31e3751d9984500a34667917c784e306
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
commit-container1:latest	cb245016a8d1	91.9MB	25.8MB	
docker/welcome-to-docker:latest	c4d56c24da4f	22.9MB	6.35MB	U
my-calc:latest	af45abf411da	91.8MB	25.8MB	U
param-demo1new:latest	532cf09e1d8f	202MB	43.5MB	U
python:latest	151ab3571dad	1.63GB	421MB	U
redis:latest	7b6fb55d8b0a	225MB	55.4MB	

```
param@Params-MacBook-Air desktop % docker run -d -p 5600:80 commit-container1
0df7457a4725acfe2282f0a7dede803cb6b215d311ba2c3b17afc0a806674eb0
param@Params-MacBook-Air desktop % docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAM
0df7457a4725	commit-container1	"/docker-entrypoint..."	22 seconds ago	Up 22 seconds	0.0.0.0:5600->80/tcp, [::]:5600->80/tcp	vig
0889c9794bde	my-calc	"/docker-entrypoint..."	22 minutes ago	Up 22 minutes	0.0.0.0:5500->80/tcp, [::]:5500->80/tcp	heu

```
ristic davinci
```

Q. Explain the difference between using `docker commit` and creating images using a `Dockerfile`, and mention when `docker commit` should be avoided.

docker commit creates an image from a running container's current state, but it is not reproducible or version-controlled.

A **Dockerfile** defines step-by-step instructions to build an image, making it consistent, reusable, and team-friendly.

Param Dholakia

docker commit should be avoided in production or team projects where reproducibility and maintainability are important and should only be implemented when doing experiments or debugging.

Container registry

Log in to Docker Hub from the command line.

→ docker login -u paramdholakia

```
param@Params-MacBook-Air desktop % docker login -u paramdholakia

Info → A Personal Access Token (PAT) can be used instead.
       To create a PAT, visit https://app.docker.com/settings

Password:
Login Succeeded
param@Params-MacBook-Air desktop %
```

Push the newly created or tagged image to Docker Hub.

→ docker push paramdholakia/my-calc

```
param@Params-MacBook-Air desktop % docker tag my-calc paramdholakia/my-calc
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
commit-container1:latest	cb245016a8d1	91.9MB	25.8MB	U
docker/welcome-to-docker:latest	c4d56c24da4f	22.9MB	6.35MB	U
my-calc:latest	af45abf411da	91.8MB	25.8MB	U
param-demo1new:latest	532cf09e1d8f	202MB	43.5MB	U
paramdholakia/my-calc:latest	af45abf411da	91.8MB	25.8MB	U
python:latest	151ab3571dad	1.63GB	421MB	U
redis:latest	7b6fb55d8b0a	225MB	55.4MB	U

```
param@Params-MacBook-Air desktop % docker push paramdholakia/my-calc
Using default tag: latest
The push refers to repository [docker.io/paramdholakia/my-calc]
d8ad8cd72600: Pushed
33257afe2855: Pushed
88799a707571: Pushed
e46c32e79d26: Pushed
821790ca706f: Pushed
31d394b0c9ed: Pushed
da8475fa07c7: Pushed
9084d2ffc283: Pushed
7833e4e4252c: Pushed
c5ad07fbd6e6: Pushed
69104fb3672f: Pushed
latest: digest: sha256:af45abf411da690f416d45f46541fde7abd16cd1c1849cc722d84e9d3f4226e4 size: 856
param@Params-MacBook-Air desktop %
```

Pull the same image back from the registry to verify it was uploaded successfully.

- Deleted all containers

```
param@Params-MacBook-Air desktop % docker rm -f $(docker ps -aq)
0df7457a4725
d889c9794bde
8b2982ed89d1
734f560abf1c
996d815353bf
0d629a55a628
a5dd5e36b237
```


Param Dholakia

- Deleted all images

```
param@Params-MacBook-Air desktop % docker rmi -f $(docker images -aq)
Untagged: commit-container1:latest
Deleted: sha256:cb245016a8d1a9f8298f9171771384df31e3751d9984500a34667917c784e306
Untagged: my-calc:latest
Deleted: sha256:af45abf411da690f416d45f46541fde7abd16cd1c1849cc722d84e9d3f4226e4
Untagged: param-demo1new:latest
Deleted: sha256:532cf09e1d8f7d3a727a54fce58ff1ea049ba4b2125d83a689b1b8f0eab59d95
Untagged: redis:latest
Deleted: sha256:7b6fb55d8b0adcd77269dc52b3cffe5f59ca5d43dec3c90dbe18aacce7942e1
Untagged: python:latest
Deleted: sha256:151ab3571dad616bb031052e86411e2165295c7f67ef27206852203e854bcd12
Untagged: docker/welcome-to-docker:latest
Deleted: sha256:c4d56c24da4f009ecf8352146b43497fe78953edb4c679b841732beb97e588b0
```

- Checking images if any exists

```
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
param@Params-MacBook-Air desktop % docker pull paramdholakia/my-calc				

- docker pull paramdholakia/my-calc

```
param@Params-MacBook-Air desktop % docker pull paramdholakia/my-calc
Using default tag: latest
latest: Pulling from paramdholakia/my-calc
69104fb3672f: Pull complete
33257afe2855: Pull complete
e46c32e79d26: Download complete
Digest: sha256:af45abf411da690f416d45f46541fde7abd16cd1c1849cc722d84e9d3f4226e4
Status: Downloaded newer image for paramdholakia/my-calc:latest
docker.io/paramdholakia/my-calc:latest
```

- rechecking images

```
param@Params-MacBook-Air desktop % docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
paramdholakia/my-calc:latest	af45abf411da	91.8MB	25.8MB	
param@Params-MacBook-Air desktop %				